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# G H Raisoni International Skill Tech University, Pune

**S.Y B. Tech (TERM-IV)**

**CAE II- Summer 2026 (2025 Pattern)**

## Theory of Computation (25BTPCL404)

**(Time: - 01 Hour)** **(Max. Marks: 20)**

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**Instructions to the students:**

*i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.*
*iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed*
*v) Assume suitable data, if necessary.*

| Q.No | Question | Marks | CO | BL |
|---|---|---|---|---|
| Q1.(a) | What is a derivation in CFG? Explain the difference between leftmost derivation and rightmost derivation. | [5] | 3 | 2 |
| b) | Consider the Grammar as given below G = (V, T, P,S) where V={S), T={0,1}, S={S} and P= { S→0S1S / 1S0S/ Ɛ} Derive string : **"011100"** with the leftmost derivation and draw Parse Tree. | [5] | 3 | 2 |
| | OR | | | |
| Q 2.(a) | Explain the Chomsky Hierarchy with neat diagram. | [5] | 3 | 2 |
| (b) | Define Grammar, Parse Tree, Ambiguous Grammar, Unambiguous Grammar with example. | [5] | 3 | 2 |
| Q 3.(a) | Compare Finite Automata (FA) and Push Down Automata (PDA). Give Five differences. | [5] | 4 | 3 |
| (b) | Explain block diagram of Push Down Automata (PDA) with its working. | [5] | 4 | 2 |
| | OR | | | |
| Q. 4 (a) | Design a PDA for the language $L = \{ a^n b^n \mid n \geq 1 \}$ and explain its working. | [5] | 4 | 3 |
| (b) | Define PDA and acceptance of string by PDA with empty stack. | [5] | 4 | 2 |